

Younes Tatari

 Personal Website —  younes.tatari@sci.utah.edu —  LinkedIn —  Google Scholar —  YouTube

Education

University of Utah

2023-Present (anticipate graduation : March 2027)

PhD in Mechanical Engineering

Thesis: Computational Modeling of Fluid-Particle Interactions: Multiphysics and Data-driven Approaches

Harbin Institute of Technology

2018-2020

MS in Energy Science and Engineering

Thesis: Numerical modeling of corona discharge using Finite Volume Method (FVM) in OpenFOAM

Skills

CFD Software OpenFOAM, SimVascular, Ansys-Fluent, Star-CCM

Fluid Physics Biotransport, Multiphase, Lagrangian particle tracking, Heat transfer, Blood flow simulation

Particle Method LIGGGHTS (CFDEM), FlowVC, DPM, MPPIC

Languages Python, Matlab, C++, git

Visualization ParaView, 3D Slicer

Courses FEM, Scientific Machine learning, Fluid Mechanics, Neural Networks

HPC Linux, SLURM

CAD Autodesk Inventor, SolidWorks

ML PyTorch, scikit-learn, TensorFlow, CNNs, PINNs, NumPy, Pandas, SciPy, Streamlit, Agentic AI

Experience

CNH Industrial

May 2026 – Aug 2026

Soil and Crop Modeling Intern

New Holland, PA, USA

- Developed a data-driven reduced-order surrogate model (ROM) from CFD simulations, using Proper Orthogonal Decomposition (POD) for dimensionality reduction and Gaussian Process Regression (GPR) for efficient prediction.

Scientific Computing and Imaging Institute at the University of Utah

Aug 2023 – present

Graduate Research Assistant

Salt Lake City, UT, USA

- Designed a novel microcatheter for targeted embolization. My design improved delivery efficiency by 10–20% compared to the standard End-Hole Catheter (SEHC) and other commercial designs. ***Provisional Patent***
- Patient Specific Modeling of Hemodynamics During Splenic Artery Embolization. 
- CFD simulation of silk-elastinlike protein polymers (SELPs) as liquid embolic agents using Non-Newtonian models in RheoTool within OpenFOAM
- Implemented four-way fluid-particle coupling in successively bifurcating vessels (CFD-DEM). 
- Combined Sparse Identification of Nonlinear Dynamical Systems (SINDy) with Stochastic Gradient Descent to overcome SINDy's limitation regarding prior knowledge of candidate terms in the library (ADAM-SINDy).
- Developed data-driven and ML to generate interpretable predictive models for fracture sealing based on CFD-DEM simulation data in geothermal systems. 

Harbin Institute of Technology

Aug 2018 – Jun 2020

Graduate Research Assistant

Harbin, China

- Investigation of cavitation under the high electric pulse in compressible fluid and electric force calculation.
- Numerical modeling of corona discharge using Finite Volume Method (FVM) in OpenFOAM.
- Facial Keypoints Detection using convolutional neural network and machine learning technics. 

Publications

- A Dual-Lumen Microcatheter for Minimizing Particle Reflux During Embolization: Proof-of-concept with multiphysics simulations 2026
Computers in Biology and Medicine 
- Investigation of particle transport in geothermal systems using integrated CFD-DEM and data-driven approaches 2026
Journal of Geothermics 
- Adam-sindy: An efficient optimization framework for parameterized nonlinear dynamical system identification 2026
Journal of Physical Review Research 
- Optimizing distal and proximal splenic artery embolization with patient-specific computational fluid dynamics 2024
Journal of Biomechanics 